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Roll No

AU-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Robotics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain Robot anatomy with diagram.
b) Explain Robot element configuration with diagram.
2. a) Discuss the difference between polar arm and articulated arm configurations.
b) What is an industrial robot? Explain classification of Industrial Robots.
3. a) Explain basic requirements of sensors in robot.
b) Explain with diagram image processing and image acquisition system used in robotics.
4. a) Briefly explain the following:
 - i) Vision sensor
 - ii) Voice sensor
 - iii) LVDT
b) Explain robot drive system and its types.

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5. Discuss following sensors and their function:- pneumatic position sensors, Range sensors, proximity sensors.
6. Discuss following
 - a) Forward kinematics and reverse kinematics of manipulators with two, three degrees of freedom
 - b) Tech pendant programming and lead through programming.
7.
 - a) What is the importance of work envelope when considering robot design?
 - b) What is program algorithm? Differentiate between walk through and lead through.
8. Write short notes on:
 - a) Non industrial application
 - b) Wrist and slip sensors
